

GOOGLE PLAY STORE ANALYTICS DASHBOARD REPORT

1. INTRODUCTION

This project presents the development of an interactive analytics dashboard based on Google Play Store data. The objective of the project is to analyze application performance, user engagement, and market trends using data visualization techniques.

The dashboard is built using Streamlit and Plotly, enabling real-time interaction, filtering, and visualization of key metrics such as installs, ratings, reviews, and revenue. The project simulates a real-world business intelligence dashboard similar to tools like Power BI or Tableau.

2. OBJECTIVES

The main objectives of this project are:

- To analyze relationships between app size, ratings, and installs
- To identify top-performing app categories globally
- To study trends in installs over time
- To compare free vs paid app performance
- To build an interactive dashboard with filters and KPIs
- To implement time-based visibility for dashboard components

3. DATASET DESCRIPTION

The project uses two datasets:

Apps Dataset (Play Store Data)

This dataset contains information about applications available on the Google Play Store.

Key Attributes:

- App Name
- Category
- Rating

- Reviews
- Size
- Installs
- Price
- Content Rating
- Android Version
- Last Updated

Reviews Dataset (Users Review)

This dataset contains user reviews and sentiment analysis data.

Key Attributes:

- App Name
- Translated Review
- Sentiment Polarity
- Sentiment Subjectivity

4. DATA PREPROCESSING

4.1 Data Cleaning

- Removed special characters such as + and , from Installs column
- Converted data types:
 - Installs → Numeric
 - Reviews → Numeric
 - Price → Numeric
- Handled missing values using errors='coerce'

4.3 Feature Engineering

New features were created:

- Revenue = Price × Installs
- YearMonth extracted from Last Updated
- Month for time-based filtering
- Android Version (numeric extraction)
- Size conversion (KB → MB)

5. DASHBOARD FEATURES

5.1 Interactive Features

- Sidebar filters (category, rating, installs)
- Dynamic charts (Plotly)
- Drill-down navigation using tabs
- Real-time filtering

5.2 KPI Cards

- Average Rating
- Total Installs
- Total Reviews
- Total Revenue

5.3 Time-Based Access Control

Each visualization is displayed only during specific time intervals, adding a unique constraint-based logic to the dashboard.

6. VISUALIZATIONS

6.1 Task 1: Bubble Chart

Objective:

Analyze relationship between app size, rating, and installs.

Filters Applied:

- Rating > 3.5
- Reviews > 500
- Installs > 50,000
- Sentiment Subjectivity > 0.5
- App name does not contain "S"
- Selected categories only

Special Features:

- Bubble size → Installs
- Game category highlighted in pink
- Category translation:
 - Beauty → Hindi
 - Business → Tamil
 - Dating → German

Time Restriction: 5 PM – 7 PM

6.2 Task 2: Choropleth Map

Objective:

Visualize global installs by category.

Filters Applied:

- Top 5 categories by installs
- Categories not starting with A, C, G, S

Special Features:

- Highlight categories with installs > 1 million
- Country mapping visualization

Time Restriction: 6 PM – 8 PM

Task 2: Choropleth Map (Global installs)

6.3 Task 3: Time Series Analysis

Objective:

Analyze install trends over time.

Filters Applied:

- App name not starting with X, Y, Z
- Category starts with E, C, B
- Reviews > 500

Special Features:

- Month-over-Month growth calculation
- Highlight growth > 20%
- Multi-category comparison

Time Restriction: 6 PM – 9 PM

6.4 Task 4: Area Chart

Objective:

Show cumulative installs over time.

Filters Applied:

- Rating ≥ 4.2
- Reviews > 1000
- Size between 20–80 MB
- Categories starting with T, P
- App name without numbers

Special Features:

- Highlight growth > 25%
- Category translation:
 - Travel → French
 - Productivity → Spanish
 - Photography → Japanese

Time Restriction: 4 PM – 6 PM

6.5 Task 5: Grouped Bar Chart

Objective:

Compare rating vs reviews.

Filters Applied:

- Month = January
- Size $\geq$ 10MB
- Rating $\geq$ 4.0

Special Features:

- Dual-axis visualization
- Top 10 categories

Time Restriction: 3 PM – 5 PM

6.6 Task 6: Dual Axis Chart

Objective:

Compare installs and revenue for free vs paid apps.

Filters Applied:

- Installs $>$ 10,000
- Revenue $>$ \$10,000
- Android version $>$ 4.0
- Size $>$ 15MB
- Content rating = Everyone
- App name $\leq$ 30 characters

Special Features:

- Top 3 categories
- Dual-axis comparison

Time Restriction: 1 PM – 2 PM

7. TOOLS USED

- Python
- Streamlit
- Pandas
- NumPy
- Plotly
- Scikit-learn
- NLTK

8. CONCLUSION

- High-rated apps tend to have moderate sizes
- Certain categories dominate installs globally
- Free apps generate more installs but less revenue per user
- Significant growth trends are visible in selected categories
- Time-based filtering adds dynamic user experience.

This project successfully demonstrates how data analytics and visualization can be used to derive meaningful insights from real-world datasets. The dashboard provides an interactive and user-friendly interface for exploring app performance and trends. The implementation of filters, KPIs, and time-based logic enhances the analytical capabilities, making it comparable to enterprise-level dashboards.